S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 8

Aim: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Demonstrate the concept of Semantic network on imputed predicate logic to represent the knowledge.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Understand semantic networks as a knowledge representation technique
- Implement ISA and HAS-A relationships
- Demonstrate inheritance in semantic networks
- Query knowledge programmatically using Python

THEORY:

Semantic Network

A **semantic network** is a graphical knowledge representation technique consisting of:

- **Nodes** → Concepts / Objects
- **Edges** → Relationships

Common relations:

- ISA → Inheritance
- HAS-A → Property
- CAN → Ability

It is closely related to **predicate logic representation**, where:

isa(dog, animal).
has(dog, tail).
can(bird, fly).

Relationship to Predicate Logic

A semantic network can be represented using first-order predicates:

Semantic Relation	Predicate Form
Dog is an Animal	isa(dog, animal)
Bird can Fly	can(bird, fly)
Dog has Tail	has(dog, tail)

Inheritance rule:

If

isa(X, Y) and has(Y, Z)

Then

has(X, Z)

Semantic Network Example

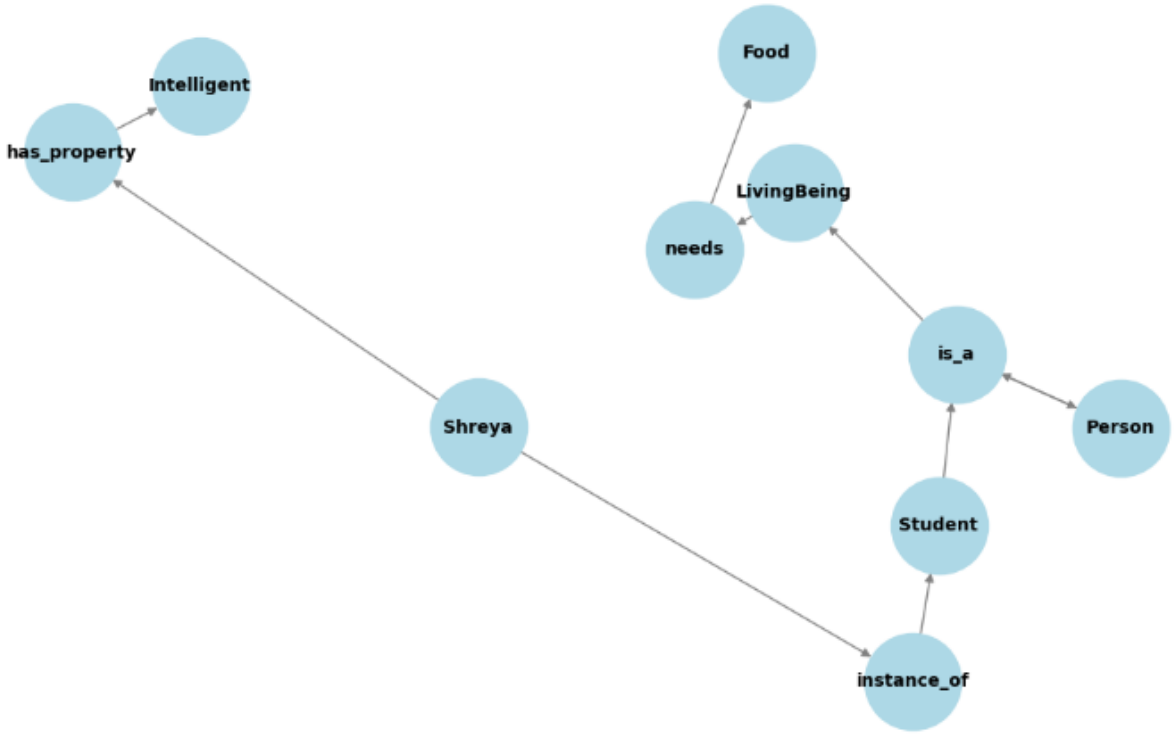
Nodes:

- Animal
- Bird
- Dog
- Sparrow

Relationships:

- Bird ISA Animal
- Dog ISA Animal
- Sparrow ISA Bird
- Animal has cells
- Bird can fly
- Dog can bark

SEMENTIC NETWORK DIAGRAM:



PROGRAM CODE:

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```python
import matplotlib.pyplot as plt
import networkx as nx

class SemanticNetwork:
    def __init__(self):
        self.network = []

    def add_fact(self, subject, relation, obj):
        # Adding the relation (subject, relation, object) as a
        self.network.append((subject, relation, obj))

    def display_network(self):
        # Displaying all the facts in the network
        if not self.network:
            print("No facts in the network.")
        else:
            print("\nCurrent Semantic Network:")
            for fact in self.network:
                print(f"{fact[0]} {fact[1]} {fact[2]}")

            # Now let's generate and display the graph of the r
            self.display_graph()

    def display_graph(self):
        # Create a directed graph
        G = nx.DiGraph()

        # Add nodes and edges to the graph based on the facts
        for fact in self.network:
            subject, relation, obj = fact
            # Create edges like: subject -> relation -> object
```

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```python
            # Create edges like: subject -> relation -> object
            G.add_edge(subject, relation)
            G.add_edge(relation, obj)

        # Draw the graph using matplotlib
        plt.figure(figsize=(10, 6))
        pos = nx.spring_layout(G)  # Positioning the nodes in the graph
        nx.draw(G, pos, with_labels=True, node_size=3000, node_color='lightblue',
                font_size=10, font_weight='bold', edge_color='gray')
        plt.title("Semantic Network Graph")
        plt.show()

def main():
    semantic_network = SemanticNetwork()

    while True:
        print("\nMenu:")
        print("1. Add a new fact")
        print("2. Display all facts and graph")
        print("3. Exit")

        choice = input("Enter your choice (1-3): ")

        if choice == '1':
            # Input the number of facts
            try:
                num_facts = int(input("How many facts do you want to enter? "))
            except ValueError:
                print("Please enter a valid number.")
                continue

            for i in range(num_facts):
```

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[3]

```python
            for i in range(num_facts):
                print(f"\nFact {i + 1}:")

                # Input Subject, Relation, Object
                subject = input("Enter subject: ")
                relation = input("Enter relation: ")
                obj = input("Enter object: ")

                # Add the fact to the semantic network
                semantic_network.add_fact(subject, relation, obj)

        elif choice == '2':
            # Display all facts and the network graph
            semantic_network.display_network()

        elif choice == '3':
            # Exit the program
            print("Exiting the program.")
            break

        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```

INPUT & OUTPUT:

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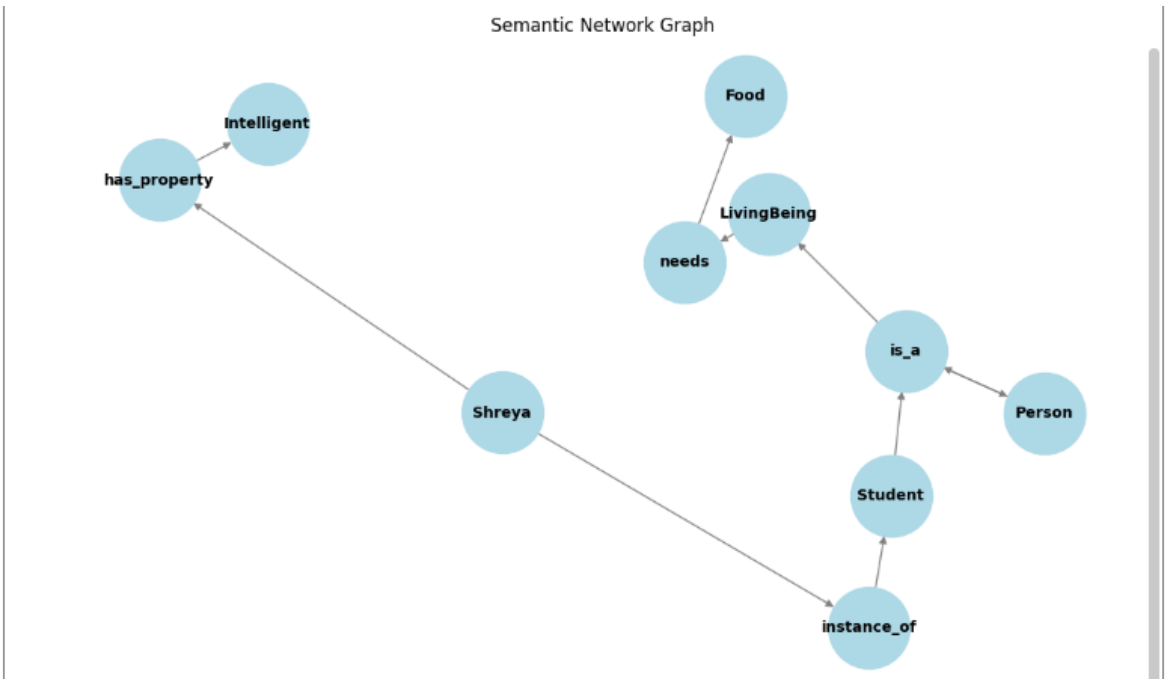
```
main()
```

```
Menu:
1. Add a new fact
2. Display all facts and graph
3. Exit
Enter your choice (1-3): 1
How many facts do you want to enter? 5

Fact 1:
Enter subject: Shreya
Enter relation: instance_of
Enter object: Student

Fact 2:
Enter subject: Student
Enter relation: is_a
Enter object: Person

Fact 3:
Enter subject: Person
Enter relation: is_a
Enter object: LivingBeing

Fact 4:
Enter subject: LivingBeing
Enter relation: needs
Enter object: Food

Fact 5:
Enter subject: Shreya
Enter relation: has_property
Enter object: Intelligent

Menu:
1. Add a new fact
2. Display all facts and graph
3. Exit
Enter your choice (1-3): 2

Current Semantic Network:
Shreya instance_of Student
Student is_a Person
Person is_a LivingBeing
LivingBeing needs Food
Shreya has_property Intelligent
```



CONCLUSION:

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DISCUSSION QUESTIONS:

1. **What is a semantic network?**

   _____

   _____

   _____

   _____

2. **What is ISA relationship?**

   _____

   _____

   _____

   _____

3. **What is inheritance in knowledge representation?**

   _____

   _____

   _____

   _____

4. **How is predicate logic used here?**

   _____

   _____

   _____

   _____

5. What are advantages of semantic networks?

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REFERENCES:

- *https://www.geeksforgeeks.org/artificial-intelligence/semantic-networks-in-artificial-intelligence/*
- *https://thecloudflare.com/knowledge-representation-using-the-semantic-network-in-ai/*
- *https://www.siliconvalleycloudit.com/knowledge-representation-in-ai-semantic-networks/*